YEDUVAKA MANOJ CHAANDU

Data Analyst | Data Scientist

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PROFESSIONAL SUMMARY

BCA Data Science student (graduating 2027) working end to end: data cleaning, exploratory analysis, modeling, and reporting a non-technical stakeholder can act on. Microsoft PL-300 certified. Projects span panel econometrics, time-series forecasting on official Indian data, Power BI dashboards, and regression models benchmarked against naive baselines.

SKILLS

Languages & Query: Python, SQL, DAX, Pandas, NumPy, Polars, Apache Arrow
Analytics: Exploratory data analysis (EDA), data cleaning, feature engineering, descriptive statistics, outlier and drift detection, KPI definition, dashboard design
Machine Learning: scikit-learn, XGBoost, regression, classification, K-Means clustering, cross-validation, hyperparameter tuning, model evaluation (R-squared, MAE, RMSE, precision, recall), SHAP explainability
BI & Visualization: Power BI (PL-300), Power Query, DAX measures, Matplotlib, Seaborn
Engineering & Tools: FastAPI, Pydantic, REST APIs, Docker, unit testing, Git, Jupyter, Excel

PROJECTS

India Air Quality Analysis (2017–2023) — Python, scikit-learn github.com/nx-manoj/Machine-learning-projects

- Consolidated **562,878 CPCB station-days** into 254,138 city-days across 248 cities, applying completeness rules and per-city medians.
- Found the network had grown twelve-fold, so rebuilt the trend on constant panels: mean AQI fell **17.9%** (2017–2023), while the seasonal cycle moves 2.56 times more.
- Benchmarked next-day forecasts against persistence: ridge on lag features cut error **8.8%**; gradient boosting added nothing.

India Agro-Power-Water Nexus — Panel OLS, XGBoost github.com/nx-manoj/Machine-learning-projects

- Assembled a **220-observation panel** of 20 states (2014–15 to 2024–25) from RBI, MoAFW, CEA and CGWB, plus a 21-year national series.
- Quantified yield drivers with panel OLS: irrigation and fertilizer add **339 and 255 kg/ha** per standard deviation; grid transmission losses subtract 264 kg/ha (all $p < 0.001$).
- Out-of-time holdout (2023–25): gradient boosting reached **R-squared 0.851** and MAE 246 kg/ha against 0.767 for pooled OLS, then segmented states into four K-Means policy regimes.

Telco Customer Churn — Power BI, Python, scikit-learn github.com/nx-manoj/Telco-Customer-Churn-IBM

- Designed a Power BI dashboard over **7,043 telecom customers** (1,869 churned) with contract and internet service slicers plus eight KPI visuals across churn drivers.
- Validated it with Python EDA, then compared Logistic Regression, Decision Tree and Random Forest classifiers to rank churn risk and derive retention actions.

Bengaluru Residential Price Analysis — Python, scikit-learn github.com/nx-manoj/Machine-learning-projects

- Modeled **13,320 property listings**, parsing area stored as text across six units and removing 14% of rows under plausibility rules; compared five models under cross-validation.
- Reached **R-squared 0.68** on held-out data with ₹27.6 lakh mean absolute error against a ₹56.0 lakh baseline; quantified a 4.8 times locality price spread and reported where the model under-predicts.

Also published: DataPilot — MIT-licensed Python EDA library on Polars and Arrow with five LLM providers (github.com/nx-manoj/DataPilot); Loan Risk Platform — XGBoost with SHAP explainability served via FastAPI and Docker (github.com/nx-manoj/loan-risk-platform).

EDUCATION

Bachelor of Computer Applications (BCA), Data Science Expected 2027
Aditya Educational Institutions, Visakhapatnam CGPA: 8.46 / 10

CERTIFICATIONS

- Power BI Data Analyst Associate (PL-300) — Microsoft